

Delivery problem

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Business Context

A restaurant chain is preparing to open a new branch in Bogotá. A significant share of the branch's expected sales will come from delivery orders, making the number of delivery drivers one of the most important operating decisions before opening.

Hiring too few drivers may generate order accumulation, longer waiting times, and deterioration in customer service. However, hiring more drivers than necessary increases labor costs and leaves capacity unused during periods of lower demand.

The company currently operates several branches with similar characteristics and has collected historical information about the arrival and fulfillment of delivery orders.

Management believes that the historical behavior of these branches can be used to estimate the expected demand and operational capacity of the new location.

Available Data

The available dataset ([click here](#)) contains the following variables:

- **order_id**: unique identifier of the delivery order.
- **branch**: branch where the order was received.
- **driver_id**: identifier of the delivery driver assigned to the order.
- **received_datetime**: time at which the order was received by the branch.
- **assigned_datetime**: time at which a delivery driver was assigned to the order.
- **completed_datetime**: time at which the driver completed the service and became available for another order.

The restaurant operates between 11:00 a.m. and 10:00 p.m.

Because the intensity of delivery orders changes throughout the day, management divides operations into three time slots:

Time Slot	Period
H1	11:00–15:00
H2	15:00–18:00
H3	18:00–22:00

For planning purposes, assume that within each time slot the arrival rate can be represented by an approximately constant rate. Management wants the branch to have sufficient capacity to cover approximately 95% of expected demand scenarios.

The new branch may hire delivery drivers under different working arrangements.

The available alternatives are:

Modality	Schedule	Hours	Cost/hour	Daily Cost
Full-time AM	11:00–19:00	8	\$20,000	\$160,000
Full-time PM	14:00–22:00	8	\$20,000	\$160,000
Part-time Lunch	11:00–15:00	4	\$24,000	\$96,000
Part-time Dinner	18:00–22:00	4	\$26,000	\$104,000

Because the schedules overlap different demand periods, the company must determine the combination of full-time and part-time drivers that provides sufficient operational capacity while minimizing daily labor costs.

considering the aforementioned then answer:

- How should the new branch structure its full-time and part-time delivery workforce in order to maintain an adequate service level at the minimum possible labor cost?